# Chapter 4: Logistic Regression and Classification in R

Syntax Notes, Exercises, and Worked Solutions

STAT452 – Applied Statistics for Engineers and Scientists II

## Contents



## 1 What this worksheet covers

For a binary response $Y \in \{0, 1\}$, linear regression can produce values outside $[0, 1]$. Logistic regression models the log-odds: $\log\{\pi(x)/(1 - \pi(x))\} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$, where $\pi(x) = P(Y = 1 \mid x)$. R fits the coefficients by maximum likelihood with `glm(..., family=binomial)`. Exponentiating a coefficient gives an odds ratio.

## 2 R syntax

```
fit <- glm(transmission ~ wt + hp, data=cars, family=binomial)
prob <- predict(fit, type="response")
pred <- ifelse(prob >= 0.5, "manual", "automatic")
exp(coef(fit))
anova(null_fit, fit, test="Chisq")
```

The 0.5 threshold is a decision rule, not a universal law. Report a confusion matrix and metrics such as accuracy, sensitivity, and specificity, preferably on held-out data.

## 3 Exercises

**Exercise 1: Compare with linear regression.** Regress binary `am` on `wt` and `hp`. Check whether fitted values lie in $[0, 1]$.

**Exercise 2: Fit and interpret the logit model.** Fit logistic regression. Compute odds ratios and explain coefficient signs while holding the other predictor fixed.

**Exercise 3: Classify observations.** Convert probabilities to classes at threshold 0.5. Produce a confusion matrix and calculate accuracy.

**Exercise 4: Threshold trade-offs.** Compare thresholds 0.3 and 0.5. Calculate sensitivity and specificity.

**Exercise 5: Validate predictions.** Split into training and test observations. Fit on training data and evaluate on test data. Explain optimistic training accuracy.

**Exercise 6: Compare models.** Use a likelihood-ratio test and AIC to compare intercept-only and two-predictor models.

# 4 Worked solutions

**Solution 1.** Linear regression is not guaranteed to respect probability bounds; inspect `range(predict(linear_bina` This motivates the logistic link. ☐

**Solution 2.** Use `exp(coef(fit))`. An odds ratio above one raises the odds of the modeled success class; below one lowers the odds, conditional on other predictors. ☐

**Solution 3.** A confusion matrix separates correct and incorrect classifications. Accuracy is $(TP + TN)/n$, but it can hide unequal costs or class imbalance. ☐

**Solution 4.** Lowering the threshold generally increases sensitivity and decreases specificity. Choose it based on the cost of false negatives versus false positives. ☐

**Solution 5.** Test observations were not used during estimation, so test performance better reflects generalization. With 32 cars, treat this split as practice rather than a definitive performance claim. ☐

**Solution 6.** The likelihood-ratio test compares nested models using change in deviance. AIC balances likelihood and parameter count; lower AIC is preferred among models compared. ☐

# Checklist

`glm -> predict(type=response) -> threshold -> confusion matrix -> held-out validation`.